PRITHVI NARAYAN CAMPUS

Tribhuvan University

Institute of Science and Technology Pokhara



An Internship Report On

"Frontend Developer Intern" At

Swoyef Labs

Submitted To:

Department of Research Committee

BSc. CSIT Program Prithvi Narayan Campus

In partial fulfilment of the requirements for the Degree in Computer Science and Information Technology

Submitted By:

Purnima Wagle [30059/078]

March 2026

GOODEX LOGISTICS PVT. LTD.

Jan 20, 2026

MENTOR'S RECOMMENDATION

I am pleased to recommend the internship report entitled "**An Intern Report on Backend Developer**," prepared under my supervision by Ms. Amisha Gurung, a dedicated student of **B.Sc. CSIT (2078)** at **Prithivi Narayan Campus, Pokhara.**

This report has been submitted in partial fulfillment of the requirements for the completion of her bachelor's degree under Tribhuvan University. I appreciate her sincere effort, commitment, and professionalism demonstrated throughout the internship period and in the preparation of this document.

Regards,
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SUPERVISOR'S RECOMMENDATION

I hereby recommend this internship report prepared under my supervision by Purnima Wagle , a B.Sc. CSIT student at Prithvi Narayan Campus, in partial fulfilment of the requirements for the B.Sc. CSIT degree. During her internship, I observed her consistent dedication and steady progress in all assigned tasks. Her work meets the academic and professional standards set by the Institute of Science and Technology at Tribhuvan University. I confirm that she has satisfactorily completed her internship under my guidance and recommend this report for final evaluation.

_____

Mr. Devilal Timilsina
Internship Supervisor
B.Sc. CSIT Program

Prithvi Narayan Campus

Tribhuvan University

Institute of Science and Technology

LETTER OF APPROVAL

This is to certify that the internship report submitted by Ms. Purnima Wagle, a student of Bachelor of Science in Computer Science and Information Technology program at Prithvi Narayan Campus, has been evaluated in accordance with the guidelines of the Institute of Science and Technology, Tribhuvan University, for partial fulfilment of the B.Sc. CSIT degree.

The report meets the required academic and professional standards and is approved for final evaluation.

Mr. Chudamani Subedi **Program Director** **Prithvi Narayan Campus**	**Mr. Devilal Timilsina** **Internship Supervisor** **Prithvi Narayan Campus**
Bal Krishna Subedi **External Examiner**	

Prithvi Narayan Campus

Tribhuvan University

Institute of Science and Technology



STUDENTS DECLARATION

I hereby declare that the internship report on project entitled "Frontend Developer Intern" submitted to the Institute of Science and Technology, Tribhuvan University, in partial fulfilment of the requirements for the degree of Bachelors of Science in Computer Science and Information Technology (B.Sc. CSIT) is a collaborative work undertaken during my internship at Swoyef Labs.

This report has not been submitted to any other institution for the award of any degree or diploma. I have properly acknowledged and cited all the sources of information used in this report in accordance with the APA reference standard.

I further declare that tasks presented in this report are genuine and has been completed during the internship period at Swoyef Labs.

Submitted By:

Purnima Wagle [30059/078]

B.Sc. CSIT, 2078 Batch

Prithvi Narayan Campus, Pokhara

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to each individua whose support and collaboration have made significant differences, either directly or indirectly, in the compilation of this report.

I would like to extend my heartfelt thanks to Swoyef Labs for providing me with the opportunity to gain practical knowledge and real-world industry experience. I am also deeply grateful to my internship mentor Mr. Sanjaya Subedi for his continuous guidance, support and valuable suggestions throughout the internship period. His professional insights and encouragement greatly helped me in successfully completing my assigned tasks.

I would also like to express my sincere appreciation to my internship supervisor, Mr. Dev Timilsina for his constant supervision, feedback and academic guidance during preparation of this report.

Furthermore, I would like to thank all of my colleagues at Swoyef Labs who directly or indirectly supported me during my internship period.

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B.Sc. CSIT 8th Semester
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ABSTRACT

This report presents the experience and learning outcomes gained during a frontend development internship at Swoyef Labs Pvt. Ltd. The internship provided practical exposure to real-world web development by allowing the application of academic knowledge in designing and improving a live business website. Major activities included developing responsive user interfaces, implementing frontend components, integrating features, and improving usability and performance using modern technologies such as React, Next.js, HTML, CSS, and Tailwind CSS. The internship also introduced professional development practices including version control, collaborative teamwork, debugging, and testing across different devices and browsers. Through hands-on involvement, technical skills, problemsolving ability, and understanding of modern frontend workflows were significantly enhanced. Overall, the internship strengthened practical knowledge and prepared the intern for future professional work in web and software development.

Keywords: *Frontend Development, React, Responsive Web Design, UI/UX Design, Web Applications, API Integration, Version Control, Backend ,Layouts.*

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List of abbreviations

API – Application Programming Interface

CRUD – Create, Read, Update, Delete

IMS - Inventory Management System

JSON -JavaScript Object Notation

JWT- JSON Web Token

NPT- Nepal Time Zone

REST- Representational State Transfer

CHAPTER 1: INTRODUCTION

1.1 Introduction to Internship

An internship is a unique educational program that aims to integrate study with planned and performance related experience. It is usually designed for the benefits of young unemployed graduates and post graduates all over the world who have completed fourteen to sixteen years of education. The major purpose of the Internship program is to develop and strengthen student's skills and to prepare them for the profession (Parveen, 2012). For example, the B.Sc. CSIT program at Tribhuvan university requires every final year student to complete a six-credit internship; normally for minimum of ten weeks for 180 working hours. This organization-based experience is designed as practical training, similar to an apprenticeship.

As a final year B.Sc. CSIT at Tribhuvan University, I completed my internship at Swoyef Labs in the Frontend Department. I was given the opportunity to work and connect my theoretical learning with practical application. The objectives of an internship also vary according to organizations and fields. Some internships focus on providing general exposure to particular industry. Others provide some specific learnings, such as developing technical skills, or giving contribution to a specific project (NACE, 2022).

1.2 Problem Statement

In college, students primarily receive theoretical knowledge, which helps them understand core concepts and foundational principles. However, this often falls short in providing real-world practical experience. In professional settings, projects involve real users, deadlines, teamwork, and unexpected challenges that are very different from classroom assignments. As a result, applying theoretical concepts to real projects and adapting to a professional environment can be challenging for many students. To bridge this gap, students must seek internships where they can gain hands-on experience, work on real systems, collaborate with industry professionals, and understand practical workflows. Through internships, they not only strengthen their technical abilities but also develop essential professional skills such as communication, teamwork, adaptability, and problem-solving. However, securing such opportunities is not easy. The technology industry is highly competitive, and many companies prefer candidates who already have practical exposure. This creates a cycle where students need experience to get internships, yet they need internships to gain that experience. Therefore, finding the right internship becomes essential for skill development, career growth, and preparing for a successful role in the job market.

1.3 Objective

The following are the internship objectives that were to be achieved:

- **Gain Practical Industry Experience**: Develop hands-on skills by working on real-world web development tasks using React, bridging the gap between classroom knowledge and professional practice.
- **Enhance Technical and Design Capabilities**: Improve front-end development proficiency by implementing clean code, responsive UI/UX designs, and modern tools, ensuring applications are functional, efficient, and user-friendly.
- **Build Professional Insight and Career Growth**: Understand workplace culture, collaborate with experienced developers, and gain exposure to industry standards, helping shape a clear direction for future career development.
- To strengthen problem-solving and debugging skills while maintaining efficient, maintainable, and scalable frontend code.

1.4 Scope and Limitations

1.4.1 Scope

The internship at Swoyef Labs involves good learning experience in backend development and digital banking. The scope includes:

- Skill Enhancement: Apply HTML, CSS, JavaScript, and frameworks in real projects
- Design Implementation: Convert UI/UX designs into responsive and interactive web pages
- API Integration: Connect frontend with backend services and handle real data.
- Responsive Development: Build websites that work smoothly on mobile, tablet, and desktop.
- Team Collaboration :Work with designers, backend developers, and project managers.
- Portfolio Development: Gain real project experience for future career opportunities.

1.4.2 Limitations

The drawbacks faced during the internship are listed below:

- Limited 10-week internship duration affecting comprehensive feature

development.

- Limited access to production-level server configurations and deployment settings due to organizational security policies.
- Balancing academic duties with internship responsibilities.
- Project requirements and timelines sometimes constrained experimentation with alternative design approaches or new technologies.

1.5 Report Organization

This report has been organized into 4 chapters:

- **Chapter 1** outlines the overview, objectives, and structure of the report.
- **Chapter 2** details the organization, its hierarchy, and domain of work.
- **Chapter 3** provides a comprehensive account of internship activities and technical involvement.
- **Chapter 4** concludes the report and summarizes learning outcomes.

References: At the end of the document, a list of references is included in APA format, ensuring proper citation.

CHAPTER 2: ORGANIZATION DETAILS AND LITERATURE REVIEW

2.1 Introduction to organization

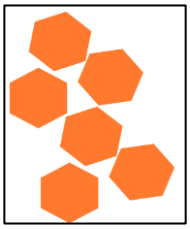


Figure 1: Organization's Logo

Swoyef Labs is a growing technology company specializing in mobile apps, game, and web development. Founded in 2023, the organization is driven by a mission to build digital products that are both functional and enjoyable to use. By combining creativity with modern technologies, Swoyef Labs focuses on delivering responsive web solutions, intuitive mobile interfaces, and engaging digital experiences. The company emphasizes user-centered design, ensuring each product is carefully crafted to meet real-world needs while maintaining high standards of performance and quality. Guided by core values such as innovation, collaboration, and excellence, Swoyef Labs encourages its team to explore new ideas, embrace challenges, and continuously improve. Since its establishment, the company has experienced steady growth, launching its first mobile application with over 10,000 downloads in the first month and expanding its team of developers and designers. Today, Swoyef Labs continues to create impactful mobile apps, interactive games, and modern web platforms, positioning itself as a rising player in the digital technology landscape.

Table 1: Organizational Contact Details

Organization Name:	Swoyef Labs Pvt. Ltd
Organizational Type:	Private Limited
Location:	Rupa-2, Kaski
Email Address:	contact@swoyef.com
Website:	https://www.swoyef.com/

2.2 Organizational Hierarchy

Swoyef Labs is led by a strong top-level management team that sets the company's vision and strategy. Under their guidance, the Project and Product Management team oversees all development work and keeps projects aligned. The Development Department is organized into several specialized teams:

Frontend team builds the user interface and ensures the application is responsive and easy to use . Backend team handles server-side logic, databases, APIs, security, and the core business processes. Design / UI-UX Team works on user experience and interface design, and collaborates with the frontend team to bring the designs to life. Quality Assurance tests the system to make sure everything is reliable and performs well.

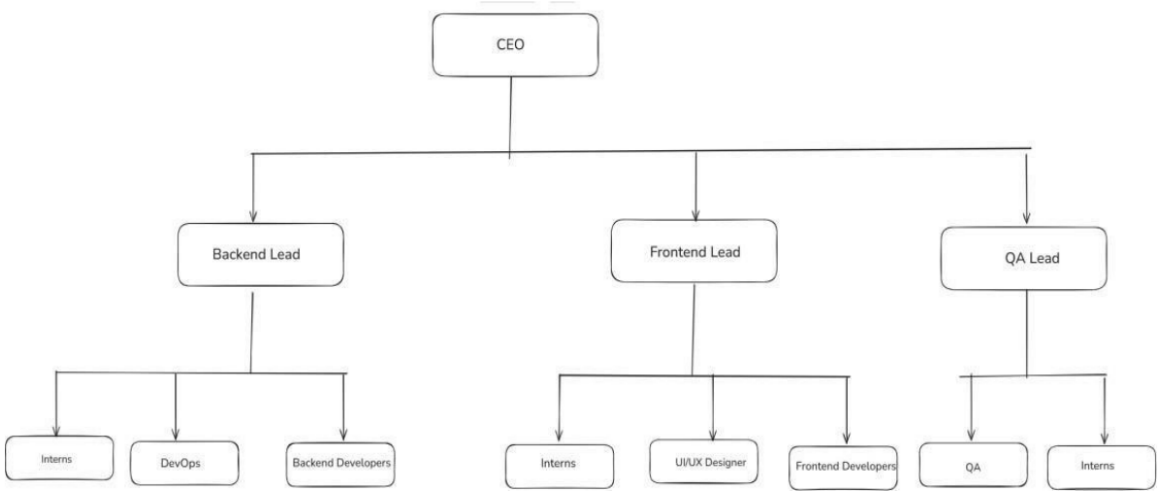


Figure 1: Organizational Hierarchy

2.3 Working Domain of Organization

Swoyef Labs Pvt. Ltd. operates in various domains of information technology, focusing on delivering modern digital solutions and software services to clients. The organization specializes in developing web-based applications and business platforms that improve operational efficiency and digital presence for organizations. The company follows modern software development practices and emphasizes quality, usability, and performance in all its projects:

Key working domains for this organization includes:

- Mobile app development
- Web app development.
- Game Development.
- Website Maintenance and Optimization.
- Quality Assurance and Testing.

2.4 Description of Intern Department/Unit

I was selected as an intern under the Frontend Development Unit of Tech Department at Swoyef Labs , following a brief interview with the HR and CTO. I worked alongside mentor, testers, developers, designers and QA team. The internship details are shown below:

Table 2: Internship Details

Office Hour	11:00 am – 6:00pm
Working Hour	7 Hours per day
Working Days	6 days a week
Mentor	Sanjaya Subedi
Total Duration	12 weeks

2.5 Literature Review

Frontend web development represents a core discipline within contemporary information technology, concentrating on the creation and implementation of graphical user interfaces that enable interaction between users and digital systems. Academic discussions in this field consistently emphasize usability, accessibility, responsiveness, and performance as essential elements in delivering effective web applications (Abras et al, 2004, 2018). As internet access expands across diverse devices and screen sizes, development methodologies have increasingly shifted toward user-centered and adaptive design principles (al, 2004) A web application can be described as a browser-based software solution that combines visual interface elements, business logic, and server-side communication to provide interactive functionality. Modern systems depend on the seamless coordination between frontend and backend technologies to deliver dynamic user experiences. Research suggests that the adoption of structured development frameworks plays a crucial role in enhancing scalability, maintainability, and efficiency in application development). These frameworks offer predefined architectural patterns, reusable components, and standardized development workflows, thereby reducing complexity and improving code organization. 14 Responsive Web Design has emerged as a foundational concept in front-end engineering (Frain, 2022). This design approach enables web interfaces to automatically adjust their layout and content presentation according to the device's screen size and resolution. Empirical studies demonstrate that responsive design significantly enhances accessibility and ensures a consistent user experience across desktops, tablets, and mobile devices . The implementation of fluid grid systems, flexible media elements, and CSS media queries allows applications to adapt dynamically without requiring separate device- specific versions. Given the rapid growth of mobile internet consumption, responsiveness is no longer optional but a standard expectation. The significance of User Interface (UI) and User Experience (UX) design is also widely documented in scholarly literature. Effective interface structuring, intuitive navigation, and visual coherence directly influence user satisfaction and engagement levels.Consequently, frontend development practices increasingly prioritize accessibility standards, and interactions. The evolution of JavaScript-based frameworks, particularly React and Next.js, has substantially reshaped frontend development methodologies. React promotes a componentdriven architecture that enables developers to build modular and reusable interface elements, improving maintainability and collaboration within teams. Studies highlight that such modularization contributes to

improved rendering performance and streamlined development processes Building upon React, Next.js introduces capabilities such as server-side rendering and advanced routing mechanisms, which enhance performance optimization, search engine. Contemporary research further underscores the importance of performance enhancement strategies in frontend systems. Techniques including code splitting, lazy loading, and modular rendering are widely applied to reduce load times and optimize resource utilization (Google). Additionally, architectural innovations such as micro-frontend design enable large-scale applications to be divided into independently deployable modules, promoting scalability and parallel team development. However, scholars also note thatsuch distributed architectures may increase integration complexity and require careful management.

CHAPTER 3: INTERNSHIP ACTIVITIES

3.1 Roles and Responsibilities

The internship was focused on frontend web development activities involving user interface design, implementation of web components, and improvement of website usability. My role involved assisting the development team in building and maintaining a live business web application while ensuring responsive and user-friendly design:

- Understanding and navigating the existing frontend codebase and architecture
- Learning API fundamentals and integrating RESTful endpoints
- Implementing reusable UI components and homepage sections.
- Implementing basic authentication and authorization
- Assisting in cart interface, authentication UI, and interactive website features.
- Identifying and resolving layout, styling, and responsiveness issues.
- Working with backend developers and team members to integrate frontend with application functionality.
- Refactoring and improving code quality and maintainability
- Testing webpages across browsers to ensure consistency and usability.

The overarching goal of this internship was to effectively bridge the gap between academic learning and professional software engineering by immersing myself in different industry projects. Through these responsibilities, I developed hands-on expertise in frontend technologies, strengthened my problem-solving abilities, learned to collaborate in a team setting, and built a solid foundation for a future career in frontend development particularly in domains like inventory management systems. This experience significantly boosted my confidence and practical understanding of full-cycle software development.

3.2 Weekly Log

Table 3: Weekly Logs

Week	Responsibilities	Activities Performed	Observations
1	Onboarding and understanding company process	Completed onboarding process, set up development environment (Node.js, Reactjs project, Git), understood project structure.	Gained clarity about company workflow, development standards, and project organization.
2	Understanding IMS architecture	Explored Inventory Management System modules, studied product, stock, and order management workflows, reviewed API integration patterns.	Developed understanding of system architecture and data flow between frontend and backend. Understood modular structure and real-world inventory processes.
3	Figma Design Implementation	Converted some of the figma UI design into responsive components, implemented layout structure and ensured alignment with design specifications	Translated UI/UX designs into functional components. Learned importance of responsiveness and pixel-perfect implementation.

4	API Integration for Landing Pages	Integrated backend APIs into landing page components, implemented dynamic data rendering, handled loading and error state.	Strengthened knowledge of API and asynchronous data handling. Understood proper error and loading state management
5	Dashboard UI Development and Improvement	Worked on Admin, Inventory, dashboard interfaces, structured data display components,	Enhanced component structuring skills and UI consistency practices. Gained experience in dashboard-based data visualization.
6	Dashboard API Integration	Reviewed completed modules, improved component structure, enhanced state handling, and ensured smooth workflow.	Improved understanding of state management and validation logic. Learned to handle dynamic dashboard data efficiently.
7	Testing and Bug fixes	Identified and resolved UI inconsistencies, fixed API related issues, improved error handling, and refactored components for maintainability.	Developed code optimization and refactoring skills. Understood importance of clean and reusable components.

8	Performance Optimization and improvements	Optimized frontend performance, reduced unnecessary re-renders, improved state handling	Strengthened debugging skills and. Improved attention to detail in UI and API handling.
9	Final Testing and Review	Participated in system testing, validating secure API flows, and code reviews.	Gained knowledge about performance tuning techniques and React rendering behavior. Understood importance of efficient component lifecycle management.
10	Advanced Feature Implementation	Implemented additional enhancements such as filtering, sorting, and reusable components for scalability.	Learned to design scalable frontend structures.
11	Integration Testing and System Validation	Participated in system testing, validated secure API flows, collaborated in code reviews and feedback implementation.	Understood importance of secure API communication and peer code review practices.

12	Final Review and Documentation	Conducted final testing, ensured UI consistency, prepared documentation of completed modules and improvements	Gained experience in project finalization, documentation standards, and production readiness evaluation.

3.3 Task/Activities Performed

Under the Frontend Development Team, I got engaged in following activities during my internship at Swoyef Labs:

3.3.1 Environment Setup & Initial Components

During the first week of my internship, I focused on setting up the frontend development environment and building the foundational components for the application. As the internship progressed, I worked on implementing advanced features such as filtering, sorting, and reusable components to enhance scalability. This helped me improve my problem-solving skills and develop better practices for component reusability, while also learning how to design scalable frontend structures.

I was also involved in integration testing and system validation, where I participated in testing processes, validated secure API flows, and collaborated with the team through code reviews and feedback implementation. This experience helped me understand the importance of secure API communication and peer review practices, while also strengthening my teamwork and collaborative development skills.

Throughout the internship, I worked with core technologies such as React for building component-based user interfaces, TypeScript for ensuring type safety and maintainable code, Vite as a fast development and build tool, and Tailwind CSS for efficient and consistent styling. My tasks included initializing the project using Vite, configuring TypeScript support, installing and setting up essential dependencies for state management, routing, and API integration, as well as creating reusable UI components like buttons, forms, tables, modals, and navigation bars. I also ensured consistent styling patterns across all components using Tailwind CSS.

3.3.2 Understanding Project Structure & Architecture

I explored the IMS codebase to understand its architecture and workflow, which helped me gain a deeper understanding of how large-scale frontend systems are structured. During this process, I studied the component hierarchy, folder structure, and state management patterns used in the application. I also learned about the multi-tenant architecture of the IMS and how the frontend interacts with multiple user roles. Additionally, I reviewed API endpoints and their integration points to prepare for future development tasks Furthermore, I worked on page design and layout implementation, where I focused on designing the landing page,

dashboards, and other user interface pages. I created responsive layouts for the landing page, login page, and dashboards, ensuring a seamless user experience across different devices. I implemented role-based rendering so that different users such as Admins, Inventory Managers, and Cashiers could access relevant UI components. I also ensured mobile responsiveness using Tailwind CSS and media queries, and developed dynamic forms with proper validations using React Hook Form, along with effective error handling mechanism.

3.3.3 API Integration

I learned how to effectively connect frontend components with backend APIs, which was a crucial part of the development process. During this phase, I implemented API calls using Axios for efficient data fetching and submission. I worked on integrating APIs related to inventory management, user authentication, transaction logs, and reporting features, ensuring seamless communication between the frontend and backend systems. Additionally, I handled API errors and managed loading states to maintain consistency, reliability, and a better user experience. I also mapped backend data to frontend components, ensuring accurate and dynamic display of dashboards, tables, and charts across the applications.

3.3.4 Performance Optimization

I also focused on improving the performance of the IMS application and efficiently managing global state across the system. During this phase, I optimized component rendering using techniques such as React.memo and the useCallback hook to minimize unnecessary re-renders. I implemented lazy loading for heavy components and images to enhance application performance and reduce initial load time. Additionally, I managed global state using the React Context API, enabling efficient data sharing across multiple components. To further improve performance, I reduced API call overhead by implementing caching strategies and conditional data fetching, ensuring a more responsive and optimized user experience.

CHAPTER 4: CONCLUSION AND LEARNING OUTCOMES

4.1 Conclusion

The frontend development internship at Swoyef Labs Pvt. Ltd. provided valuable practical experience by connecting academic knowledge with real-world software development practices. Throughout the internship, I was involved in designing and improving user interfaces, implementing responsive layouts, integrating frontend features, and resolving usability issues within a live web application environment. This experience allowed me to understand how modern web technologies are applied in professional projects to deliver efficient and user-friendly digital solutions. Working on real development tasks helped me gain a deeper understanding of frontend workflows, including component-based development, responsive design implementation, debugging techniques, and collaborative coding practices. Exposure to version control systems and team-based development improved my ability to manage tasks systematically and communicate effectively with other developers. The internship also highlighted the importance of usability, performance optimization, and continuous testing in delivering high-quality web applications. Through practical problem solving and regular interaction with development tools and technologies, I strengthened both my technical competence and professional confidence. This internship experience significantly enhanced my knowledge of frontend development and prepared me for future opportunities in web and software development. It provided a strong foundation for applying modern development practices and adapting to evolving technologies in the IT industry.

Overall, this internship has provided me with a strong foundation in frontend development and significantly enhanced my understanding of how software systems support supply chain and logistics operations. It has also increased my motivation and readiness to further develop my skills and pursue a professional career in frontend engineering.

4.2 Learning Outcomes of Internship Activities

Throughout the course of my twelve-week internship, I learned a lot. Learning outcomes of my internship is listed below:

- **Technical Growth**: I became much more confident with React and Typescript for building production-ready frontend designs, handling tokens. I also learned practical debugging, testing, and optimization techniques that aren't always covered deeply in classes.
- **Responsive Web Design Skills**: I learned how to design and implement responsive user interfaces that adapt to different screen sizes and devices. This improved my understanding of layout structuring, media queries, and mobile-friendly design practices
- **Professional Skills**: Working in a real team taught me agile practices daily stand-ups, code reviews, Git workflows, and communicating progress clearly. I learned to take feedback seriously, document my work properly, and manage tasks under deadlines.
- **Debugging and Problem-Solving Ability**: Through regular debugging activities, I developed the ability to identify and resolve interface issues, styling conflicts, and component rendering problems using browser developer tools and logical troubleshooting approaches.
- **Personal Development:** It boosted my confidence in contributing to live projects and showed me I'm capable of handling real-world challenges. It also motivated me to keep learning about modern frontend tools Developed confidence in reading and incrementally improving existing production codebase as a complete beginner.

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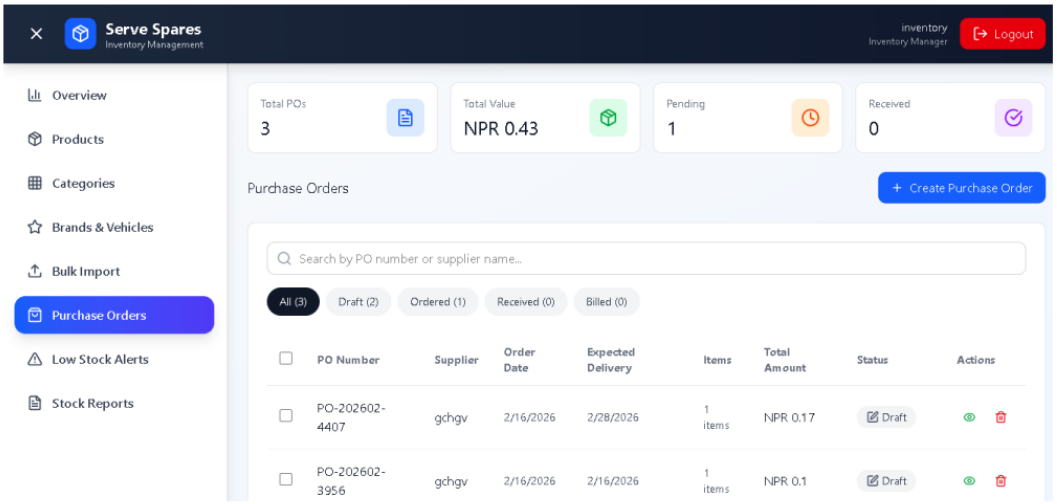
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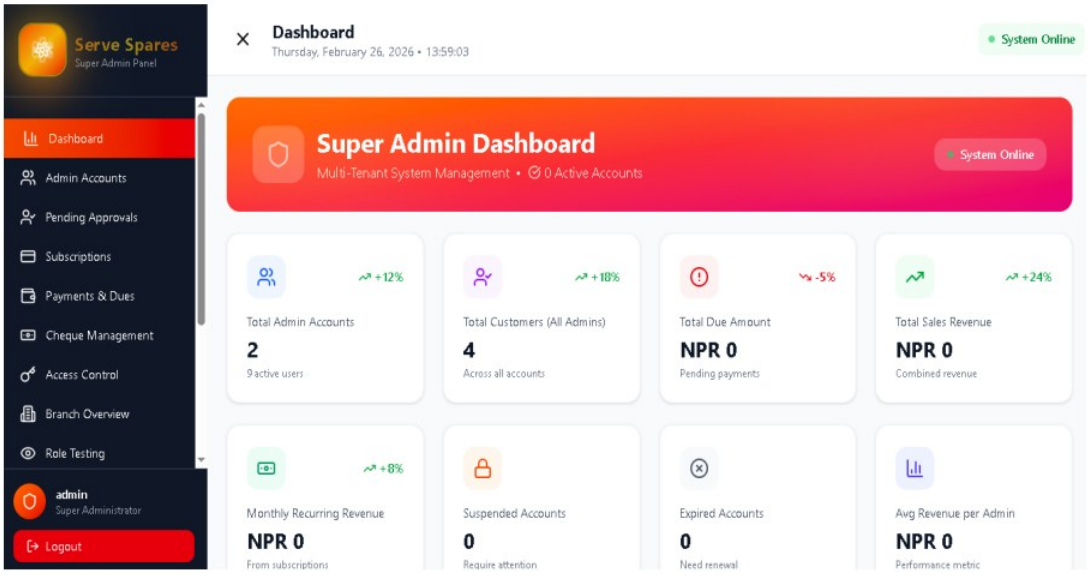
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Appendices

1.My Works





2.Workspace

